
The Architecture of Errors: Logarithmic Mode Discovery and Polylogarithmic Intervention Budgets for Long-Context LLM Reliability

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Abstract

1 Prior work on length-decay benchmarks [Anonymous, 2025] argued that LLM
2 reliability does not decay exponentially with output length because errors concen-
3 trate at sparse “key tokens” rather than spreading uniformly. This paper extends
4 that framework along an orthogonal axis: LLM errors are not only sparse, they
5 are *repetitive*. The same handful of failure patterns recur across models, datasets,
6 and domains. Inside the small set of key tokens, a latent fraction β produce *hard*
7 failures; those hard failures cluster into a finite catalogue of recurring modes whose
8 count grows slowly with the number of observed failures. Under an empirical postu-
9 late of logarithmic mode discovery—calibrated against endpoint counts from three
10 published taxonomies (ErrorAtlas, MWPEs, HumanEval-error-categorisation),
11 with a direct subsample-discovery curve still outstanding—the intervention budget
12 required to bound *per-hard-token* residual error scales polylogarithmically in se-
13 quence length. Sequence-level reliability targets are strictly tighter and approach
14 full-catalogue coverage as the number of hard decisions grows—an asymmetry we
15 surface explicitly rather than burying. Twelve intervention categories from prior
16 work, plus 28 capability-elimination citations across six axes stratified into Patterns
17 A/B/C (Appendix B), are consistent with a small library on the order of tens of
18 interventions covering the head of the failure distribution in the per-hard-token
19 regime. We analyze the polylog conclusion symbolically under several cluster-
20 count laws (logarithmic, Heaps, saturating; Appendix A) and find it robust across
21 the family, though the specific doubly-logarithmic rate is the optimistic special
22 case. Prominent steep-decay counter-evidence [Dziri et al., 2023, Kuratov et al.,
23 2024, Kwa et al., 2025], re-audited in Appendix C, decays over variables other than
24 raw token length—compositional graph size, fact count, log-time horizon. This
25 relocates rather than dissolves the practical worry: the framework identifies which
26 interventions help and which do not, but does not claim the regimes where k_{hard}
27 grows fast are thereby made easy.

28 1 Introduction

29 The standard worry about long-context autoregressive generation is exponential. If every token
30 has independent error probability e , then after n tokens the chance of a fully correct output is
31 $(1 - e)^n$, which collapses to zero for any nontrivial e and large enough n [LeCun, 2023, Dziri et al.,
32 2023]. Earlier work in this series [Anonymous, 2025] argued that this worry is misplaced because
33 errors are not distributed uniformly across tokens: only 5–10% of tokens are “key”—genuinely

dependent on long-range context [Fang et al., 2025]—while the remaining majority become nearly deterministic once enough context accumulates. Replacing $(1 - e)^n$ with the two-rate model $P(\text{correct}) = (1 - e_{\text{key}})^k(1 - e_{\text{non}})^{n-k}$ with $k \ll n$ scaling sublinearly in n recovers the observed long-context coherence and converts the reliability question from “how does n grow?” to “how does k grow?”.

From sparse tokens to recurring patterns. This paper takes the next step. Sparsity tells us *where* errors live; the natural follow-up is *what* they are. Recent failure-mode atlases supply a striking empirical answer: errors are not only sparse, they are also *repetitive*. ErrorAtlas [Ashury-Tahan et al., 2026], covering 83 models across 35 datasets, organises observed failures into 17 named categories sorted by prevalence, with a long-tailed distribution heavily concentrated in the head. In code, two error types (AssertionError and NameError) cover 86.35% of failures on HumanEval across 14 LLMs, replicated across 23 models on HumanEval Pro and MBPP Pro [Wen et al., 2024]. In math, MWPEs-300K [Sun et al., 2025] categorises 304,865 errors from 15 LLMs across four math word-problem datasets and reports that dataset characteristics shape error patterns systematically. The same picture appears in multi-hop QA [Zhang et al., 2026], agentic tool use [Cemri et al., 2025], and retrieval-augmented generation [Wood and Forbes, 2024]. These observations suggest a third layer in the reliability architecture, complementing key-token sparsity (Layer 1) and within-key stratified-manifold structure (Layer 2 in Anonymous 2025). Within the small set of key tokens, only a fraction β produce *hard* failures, and those hard failures cluster into a finite catalogue of recurring modes whose size grows much more slowly than the number of observed events.

Contributions. The central result is the conditional polylog intervention budget of Proposition 1: $m \geq \lceil |C|^{1-\varepsilon/e_{\text{hard}}} \rceil$, with a doubly-logarithmic rate as the optimistic special case under the postulate above. The *sequence*-level analogue is strictly tighter and approaches full-catalogue coverage as k grows—an asymmetry we surface rather than bury. Around this central claim the paper does four supporting things. The three-layer formal framework—sparsity (α), stratification (β), and bounded mode catalogue ($|C|$)—partitions key tokens into easy and hard subsets, then adds the recurring-pattern catalogue (§3). The empirical postulate of logarithmic mode discovery is calibrated against ErrorAtlas, HumanEval, and MWPEs, with $\sigma \in [0.87, 1.85]$ across anchors and $\sigma \approx 1.85$ adopted from ErrorAtlas as the conservative target. Section 4 synthesises evidence for clustering, cluster-selective interventions, and sublinear length scaling, drawing on ~ 60 prior published results including the six-axis capability-elimination harvest of 28 quantitatively-anchored citations stratified into Patterns A/B/C (Appendix B). And the re-audit of the most-cited steep-decay results (Appendix C) shows each decays over a variable other than raw token length.

2 Related Work

Failure-mode taxonomies. ErrorAtlas [Ashury-Tahan et al., 2026], MWPEs-300K [Sun et al., 2025], the HumanEval categorisation of Wen et al. [2024], and the RFMDataset [Guo et al., 2025] together establish that, at any given corpus size, the number of distinct named failure modes is small (typically 8–20) and the cumulative coverage of the top modes is high. Domain-specific taxonomies for multi-hop QA [Zhang et al., 2026], multi-agent systems [Cemri et al., 2025], and tool-augmented agents [Yao et al., 2024] report the same pattern.

Targeted interventions. Each named cluster has been the subject of focused intervention research. Python execution closes most of the arithmetic-error cluster [Gao et al., 2023a, Chen et al., 2023, Gou et al., 2024b]; constrained decoding eliminates format violations [Suresh et al., 2025, Wang et al., 2025b, Dong et al., 2025]; execution feedback closes most code-logic errors [Shinn et al., 2023, Huang et al., 2023]; process reward models reduce step-level reasoning errors [Wang et al., 2024b, Lightman et al., 2024]; RAG strongly reduces hallucinations [Wood and Forbes, 2024]; preference optimisation closes over-refusal [Karaman et al., 2024]; structured uncertainty fixes most tool-call failures [Suri et al., 2025]. Two structural observations recur: each intervention is *cluster-selective* (residuals belong to a different named cluster, not to the targeted one), and *additivity is approximate but not perfect* [Patel et al., 2026, Le, 2026].

Length-decay benchmarks. A parallel literature measures the *shape* of the reliability decay curve. Mild-decay results—Loong [Wang et al., 2024a], GSM- ∞ [Zhou et al., 2025], RULER [Hsieh et al.,

2024], anchor-based LLMs [Pang et al., 2024]—report log-linear, sigmoidal, or threshold-like decay inconsistent with smooth $(1 - \varepsilon)^n$. Steep-decay results [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026] are sometimes cited as exponential-compounding evidence. The apparent tension dissolves on careful reading: every steep-decay paper decays over a variable distinct from raw token length, which Appendix C documents.

Gap. What is missing is a quantitative bridge between the small, repeating taxonomy catalogue and a polylog intervention budget. We provide that bridge in §3.

3 Theoretical Framework

Levels of $|C|$. LLM error analysis routinely conflates four distinct objects, which we keep separate. *L1: failure events*—concrete observed errors. *L2: empirical taxonomy categories*—researcher-chosen labels grouping L1 events; what published taxonomies report. *L3: latent failure modes*—the unobserved underlying clusters L2 approximates. *L4: capability axes / interventions*—the engineering unit (Python interpreter, constrained decoder, RAG). Throughout this section, $|C|$ refers to the L2 count, because L2 is what published taxonomies report; Postulate 1 is engineering-relevant because L4 is coarser than L2, and epistemically conditional because L2 is a noisy proxy for L3.

3.1 β -stratification of key tokens

The two-rate model of Anonymous [2025] distinguishes k key tokens (error rate e_{key}) from $n - k$ non-key tokens ($e_{\text{non}} \ll e_{\text{key}}$). Empirical atlases suggest that even within the key-token class errors are not uniformly distributed: most key tokens are “decisions” for which the model has stable representations; only a fraction concentrate the actual failures.

Definition 1 (Hard fraction). Partition the k key tokens of a sequence into easy and hard subsets,

$$k_{\text{hard}} = \beta k, \quad k_{\text{easy}} = (1 - \beta)k, \quad \beta \in (0, 1),$$

where hard key tokens have an elevated error rate e_{hard} corresponding to manifold-transition decisions in the sense of Anonymous [2025, §3.2], and easy key tokens have $e_{\text{easy}} \approx e_{\text{non}}$.

Under Definition 1, the composed sequence-level reliability becomes

$$P(\text{correct}) = (1 - e_{\text{hard}})^{\beta k} (1 - e_{\text{easy}})^{(1-\beta)k} (1 - e_{\text{non}})^{n-k}. \quad (1)$$

We do not assume iid token errors: the three rates summarise *conditional per-decision failure hazards* after conditioning on prior decisions being correct, and grouping by stratum yields the multiplicative survival expression. Because $e_{\text{easy}} \approx e_{\text{non}}$ once context accumulates, the failure rate is dominated by the βk hard tokens, and we treat e_{hard} as the load-bearing quantity. The parameter β is *latent*: failure-mode atlases report $\Pr(\text{category} \mid \text{error occurred})$, which does not identify $\Pr(\text{hard} \mid \text{key-token decision})$. We carry β symbolically.

3.2 Empirical postulate: logarithmic mode discovery

The βk hard-token failures cluster into a finite set of recurring patterns indexed $c = 1, \dots, |C|$. The natural question is how $|C|$ grows with the number of observed failures. Two candidates exist in the type-token literature: Heaps’ law (power-law, $|C| \approx K \cdot k_{\text{hard}}^b$ with $b \in [0.4, 0.6]$ [Manning et al., 2008]), and logarithmic discovery. **Zipfian rank-frequency does not imply logarithmic growth:** Heaps’ law is the correct type-token consequence of Zipf-distributed events, and it is power-law, not logarithmic. We therefore state logarithmic mode discovery as an empirical postulate defensible by direct measurement, not as a theorem.

Two sample-size variables matter and conflating them is the seam that lets reviewers in. Let $h(n) = \beta k(n)$ denote the number of hard-token decisions in a single sequence of length n ; let T denote observed hard-failure events in a corpus used to discover the domain catalogue. We write C_D for the full reachable catalogue inside domain patch D , $C_{\text{seen}, D}(T)$ for the subset discovered after T failures, and $C_{\text{active}, D}(n)$ for the subset a single sequence of length n can activate.

129 **Postulate 1** (Patch-indexed catalogue discovery). Within a fixed application domain D , the number
 130 of named recurring failure modes discovered after T sampled hard-failure events is bounded by

$$|C_{\text{seen},D}(T)| \leq \min(A_D + \sigma_D \ln T, |C_D|), \quad \sigma_D > 0, A_D \geq 0.$$

131 The cap $|C_D|$ is the domain-imposed ceiling. The postulate concerns catalogue *discovery*, not the
 132 number of hard decisions inside a single sequence.

133 **Assumption 2** (Per-sequence active-mode exposure). For a single sequence of length n in domain D ,

$$|C_{\text{active},D}(n)| \leq \min(A'_D + \sigma'_D \ln h(n), |C_D|).$$

134 Primed constants are distinct from those of Postulate 1: corpus discovery and per-sequence activation
 135 need not share the same rate.

136 Proposition 1’s sequence-length scaling uses $C_{\text{active},D}(n)$; full domain-library budgeting uses C_D .

137 **Empirical defense.** ErrorAtlas: $k_{\text{hard}} \gtrsim 10^4$, $|C| = 17$, giving $\sigma \approx 17/\ln(10^4) \approx 1.85$ with
 138 $A = 0$. HumanEval [Wen et al., 2024]: $|C| \in [8, 12]$, $\sigma \in [0.87, 1.30]$. MWPEs: $|C| \in [15, 20]$,
 139 $\sigma \in [1.2, 1.6]$. Across the three anchors, $\sigma \in [0.87, 1.85]$; we carry $\sigma \approx 1.85$ as the conservative
 140 target. Endpoint counts at a single corpus scale are not discovery curves; the subsample-discovery
 141 measurement remains an explicit empirical test we flag (§6, Appendix A).

142 **Capabilities are coarser than error classes.** Postulate 1’s $\sigma \approx 1.85$ is calibrated against *named*
 143 *error categories*, but a single capability axis often targets several L2 categories at once. A Python
 144 interpreter removes the execution-error component of arithmetic, unit conversion, simple counting,
 145 list manipulation, and date arithmetic; a constrained decoder eliminates by construction the format-
 146 violation cluster and the structural component of “missing required element.” Postulate 1 is therefore
 147 a conservative input for the engineering question; the formal coverage problem is weighted set cover,
 148 with the ranked-category form of Proposition 1 as a tractable proxy. An alternative power-law (Heaps)
 149 formulation and symbolic-form sensitivity analysis across candidate cluster-count laws are given in
 150 Appendix A.

151 **Domain patches cap the catalogue.** A model deployed in a fixed application domain occupies
 152 a bounded sub-region of the stratified manifold—a *domain patch* P_D [Anonymous, 2025, Li and
 153 Sarwate, 2025]. Empirical support: embedding-space intrinsic dimensions of ~ 5 – 15 across domains,
 154 well below ambient $\sim 10^3$ [Park et al., 2024, Li and Sarwate, 2025]; cross-domain accuracy spreads
 155 of 20–70% on a single base model [Wang et al., 2024c]; log-popularity thresholds below which
 156 scaling fails [Mallen et al., 2023, Kandpal et al., 2023]. None of these *measure* $|C_D|$ directly; together
 157 they motivate the patch-indexed hypothesis. The patch makes logarithmic or saturating discovery
 158 dynamics plausible inside fixed domains; $\sigma_D, A_D, \beta_D, |C_D|$ are all domain-indexed.

159 3.3 Coverage by a targeted intervention library

160 Given a catalogue of $|C|$ recurring modes, an intervention library of size m covers the top- m , leaving
 161 $|C| - m$ uncovered. We adopt the log-coverage form

$$F(m; |C|) = \min\left(1, \frac{\ln m}{\ln |C|}\right)$$

162 as a *postulated* closed expression respecting the $\min(1, \cdot)$ cap. For our ErrorAtlas anchor ($|C| =$
 163 $17, m = 5$), $F(5; 17) \approx 56.8\%$ and $F(10; 17) \approx 81.3\%$. This is an empirical best-fit choice, not
 164 derived from Zipf: ErrorAtlas sorts its 17 categories by prevalence but does not publish a single
 165 cross-domain cumulative top- m curve at this writing; alternative cumulatives (Zipf–Mandelbrot,
 166 truncated power law, saturating exponential) match the same qualitative shape with comparable
 167 freedom. Appendix A shows the polylog conclusion survives across them. Two caveats: (a) F counts
 168 named L2 categories covered, not L4 capability axes deployed, so real-deployment residual error
 169 is often lower at any given m ; (b) we adopt $F(0; |C|) := 0$ and $F(1; |C|) := p_1$ (mass of the most
 170 prevalent mode) since the log form is not anchored at $m = 1$. After deploying the top- m library, the
 171 residual per-hard-token error rate is $e_{\text{res}}(m) = (1 - F(m; |C|)) e_{\text{hard}}$.

172 3.4 Main proposition: conditional polylogarithmic intervention budget

173 We compose the three ingredients with the two-rate model. For sequence-length scaling we use
 174 Assumption 2, not Postulate 1: the empirical sparsity $\alpha = k/n \in [0.05, 0.10]$ is a *finite-length*
 175 statement, and under Anonymous [2025, §3.1]’s $k \sim \log n$ regime, α decays as $O(\log n/n)$.

176 **Conditionality.** The bound below is engineering math, not a theorem about LLMs. It holds
 177 conditional on (i) the log-coverage form of §3.3 on its declared domain $m \geq 2$, $|C_{\text{eff}}| \geq 2$; (ii) the
 178 residual target $\varepsilon \in (0, e_{\text{hard}})$; (iii) Assumption 2 for any sequence-length rate claim. A different
 179 cumulative fitted to the same anchors preserves the qualitative polylog conclusion (Appendix A) but
 180 changes the exponent.

181 **Proposition 1** (Per-Hard-Token Intervention Budget). *Let $\varepsilon \in (0, e_{\text{hard}})$ be a target per-hard-token*
 182 *residual error rate for sequences in domain D , with $|C_{\text{eff}}| \geq 2$, where*

$$C_{\text{eff}} = \begin{cases} C_{\text{active}, D}(n) & (\text{per-sequence scaling}), \\ C_D & (\text{full domain-library deployment}). \end{cases}$$

183 *Under the coverage form of §3.3, the smallest intervention library satisfying $e_{\text{res}}(m) \leq \varepsilon$ has size*

$$m \geq \left\lceil |C_{\text{eff}}|^{1-\varepsilon/e_{\text{hard}}} \right\rceil. \quad (2)$$

184 *If $C_{\text{eff}} = C_{\text{active}, D}(n)$ and $k(n) = \Theta(\log n)$, then in the pre-cap regime*

$$m = O((\log \log n)^{1-\varepsilon/e_{\text{hard}}}),$$

185 *i.e., the intervention budget grows doubly logarithmically with sequence length. In the cap regime, or*
 186 *for full domain-library deployment, $m \geq \lceil |C_D|^{1-\varepsilon/e_{\text{hard}}} \rceil$ independent of n .*

187 *Proof.* From §3.3, $e_{\text{res}}(m) = (1 - F(m; |C_{\text{eff}}|)) e_{\text{hard}}$. Requiring $e_{\text{res}}(m) \leq \varepsilon$ gives $F(m; |C_{\text{eff}}|) \geq$
 188 $1 - \varepsilon/e_{\text{hard}}$. Since $\varepsilon < e_{\text{hard}}$, the required coverage lies below one and the cap in F is inactive; with
 189 $F = \ln m / \ln |C_{\text{eff}}|$, we obtain $m \geq |C_{\text{eff}}|^{1-\varepsilon/e_{\text{hard}}}$, and taking the ceiling gives (2). Assumption 2
 190 gives $|C_{\text{active}, D}(n)| = O(\log \log n)$ when $k(n) = \Theta(\log n)$, before the cap. After the cap, $|C_{\text{eff}}| =$
 191 $|C_D|$ and the bound is independent of n . $\square$

192 We label this a *Proposition* rather than a *Theorem* to keep its conditional nature visible. The doubly-
 193 logarithmic rate is the *optimistic special case*: per-sequence, pre-cap, under logarithmic mode
 194 discovery. Weaker (but still polylog) rates obtain under Heaps (Appendix A); the cap regime is a
 195 domain-constant.

196 **Sequence-level versus per-hard-token bounds.** Proposition 1 bounds the *per-hard-token* residual
 197 error. Sequence-level failure probability is strictly stronger. Define the non-hard-token survival factor

$$S_{\text{base}} = (1 - e_{\text{easy}})^{(1-\beta)k} (1 - e_{\text{non}})^{n-k}, \quad (3)$$

198 so that the composed survival from (1) can be written $P(\text{correct}) = (1 - e_{\text{res}})^{\beta k} S_{\text{base}}$. The sequence-
 199 level target $P(\text{correct}) \geq 1 - \varepsilon_{\text{seq}}$ then yields three regimes:

200 **Regime (i):** $S_{\text{base}} < 1 - \varepsilon_{\text{seq}}$. Even at $e_{\text{res}} = 0$ the surviving non-hard-token failure mass exceeds
 201 the sequence-level budget. Hard-token interventions alone cannot meet the target; this is the
 202 back-door route by which the exponential-in- n concern re-enters, and it should be diagnosed
 203 before any catalogue-budgeting exercise.

204 **Regime (ii):** $S_{\text{base}} \geq 1 - \varepsilon_{\text{seq}}$. Define

$$\tau_{\text{seq}} = 1 - \left(\frac{1 - \varepsilon_{\text{seq}}}{S_{\text{base}}} \right)^{1/(\beta k)}. \quad (4)$$

205 For $0 < \tau_{\text{seq}} < e_{\text{hard}}$, the sequence target is met whenever $e_{\text{res}}(m) \leq \tau_{\text{seq}}$, so the required library
 206 size is

$$m \geq \left\lceil |C_{\text{eff}}|^{1-\tau_{\text{seq}}/e_{\text{hard}}} \right\rceil.$$

207 As the target tightens toward the feasibility boundary, $\tau_{\text{seq}} \rightarrow 0$ and $m \rightarrow |C_{\text{eff}}|$ —essentially
 208 every named mode must be covered.

209 **Regime (iii):** $\tau_{\text{seq}} \geq e_{\text{hard}}$. The sequence target is already met at the baseline hard-token rate; no
 210 intervention is required.

211 The widely-quoted “ ≈ 50 patterns $\rightarrow \approx 90\%$ reduction” claim is a per-hard-token statement; the
 212 sequence-level analogue requires correspondingly more interventions. Production systems with one-
 213 shot reliability targets should plan for the stricter regime—and check whether they are in Regime (i),
 214 where catalogue-budgeting cannot help at all.

4 Empirical Evidence

We collect evidence for three load-bearing claims: errors cluster into a small recurring set (A); each cluster is addressable by one targeted intervention (B); reliability decays sublinearly with output length (C).

Claim A: Failure-mode clustering. The strongest single result is ErrorAtlas [Ashury-Tahan et al., 2026]: 83 models $\times$ 35 datasets, $\approx$ 7,000 failures, 17 named categories with a long-tailed, head-concentrated prevalence ordering. Domain-specific Paretos reproduce the shape inside each domain—math (MWPEs top-4 categories dominate per model with strong cross-model overlap [Sun et al., 2025]), code (AssertionError+NameError = 86.35% on HumanEval, replicated across 23 models [Wen et al., 2024]), multi-hop QA (a single dominant “missing-evidence” mode [Zhang et al., 2026]), agentic tool use (under 20 recurring modes across two studies [Cemri et al., 2025, Yao et al., 2024]), and RAG (a handful of distinguishable hallucination types [Wood and Forbes, 2024]). The catalogue is also *stable* across models: ErrorAtlas reports a fixed category hierarchy across 83 models; RFMDataset [Guo et al., 2025] finds “strikingly similar failure mode distributions” across ten advanced reasoning models; EDIT [Dai et al., 2025] measures a $\approx$ 4.7% key-step fraction stable across models. Schaeffer et al. [2025] provide the closest theoretical anchor, proving that the observed aggregate power-law scaling of LLM evaluations requires the per-problem success-rate distribution to be heavy-tailed near $p = 0$ —structurally related to Postulate 1.

Claim B: Cluster-selective capability interventions. A dedicated harvest yields 28 capability-elimination citations across six independent axes (arithmetic, code execution, format/structure, perception/grounding, knowledge/RAG, verification), each axis independently confirmed by between three and nine citations. The full table and the three structural patterns—Pattern A by-construction, B strong-empirical-with-class-shift, C moderate-with-shift—appear in Appendix B.

Arithmetic is the cleanest case: PAL [Gao et al., 2023a] lifts GSM-Hard from 20.1% to 61.5%, and the residuals are problem-comprehension errors rather than arithmetic ones. Format violations admit something stronger still—constrained decoding zeroes invalid-token probability by construction [Suresh et al., 2025, Wang et al., 2025b, Dong et al., 2025]. Code-logic errors yield to execution feedback (Reflexion + AgentCoder push HumanEval pass@1 from 80% to 96.3%, with residuals in spec-misinterpretation [Shinn et al., 2023, Huang et al., 2023]); reasoning steps to process supervision (Math-Shepherd: Mistral-7B MATH 28.6% $\rightarrow$ 43.5% [Wang et al., 2024b]); hallucinations to RAG, where Acurai [Wood and Forbes, 2024] reports 100% elimination on RAGTruth—a benchmark-conditional empirical result, not a general by-construction guarantee. Two more clusters round out the picture: POROver lifts the not-overrefused rate from 57.6% to 82.1% [Karaman et al., 2024], SAGE-Agent raises When2Call accuracy from 36.5% to 65.2% [Suri et al., 2025].

The class-shift signature—post-intervention residuals belong to structurally different classes—is the empirical content of the cluster-orthogonality claim that underwrites Proposition 1’s composition step. Capabilities are coarser than error classes: a single Python interpreter removes execution-error components from five named clusters; constrained decoding eliminates format and the structural component of “missing required element” jointly. A negative control sharpens the selectivity claim. DebugBench [Tian et al., 2024] reports that execution feedback works for syntax and reference errors and is explicitly “unhelpful for logic errors”—exactly where the framework predicts capability provisioning fails.

Claim C: Sublinear length scaling. Direct evidence for sublinear—rather than exponential—decay shows up across very different measurement designs. Loong [Wang et al., 2024a] reports GPT-4o on Chain-of-Reasoning declining 81.6% $\rightarrow$ 32.9% across the 10K $\rightarrow$ >200K context bins; log-linear over $\log L$, and decisively non-exponential per-token. GSM- ∞ [Zhou et al., 2025] attacks the independent-error model from a different angle: exponentially more inference compute yields only *linear* AUC gains, and DeepSeek-R1 maintains 10%+ accuracy at 130 reasoning operations where $(1 - \epsilon)^{130}$ for any reasonable ϵ predicts near-zero. Press et al. [2023] find a $\approx$ 40% compositionality gap that is approximately *constant* across the GPT-3 family—direct evidence against per-step independent compounding, which would predict the gap to grow with chain length.

Structural correlates of the same phenomenon appear elsewhere. Pang et al. [2024]’s Anchor LLMs achieve $\approx$ 99% K/V reduction with only minor accuracy compromise: the effective number of distinct

attention-key vectors a model needs is far smaller than n . Think-Prune-Train [Costello et al., 2025] raises Pass@1 while Pass@20 plateaus—the manifold-transition signature. RULER [Hsieh et al., 2024] finds threshold behaviour inconsistent with smooth $(1 - \varepsilon)^n$. METR [Kwa et al., 2025] fits logistic-in-log-length, which is mathematically sublinear in length itself.

Self-consistency [Wang et al., 2023] is consistent with but does not *prove* clustered structure: Concordet aggregation of iid Bernoulli chains can yield similar gain magnitudes, so we treat it as suggestive rather than diagnostic. Prominent steep-decay counter-evidence [Dziri et al., 2023, Kuratov et al., 2024, Wan et al., 2026, Karpinska et al., 2024] decays over variables distinct from raw token length; per-paper re-audits showing the relocation pattern are in Appendix C.

5 Practical Implications

Reliability engineering is local patch coverage. Within a fixed domain patch, Proposition 1 makes reliability a small-catalogue engineering problem rather than an asymptotic scaling problem. A team should initially budget for a library on the order of tens of interventions, then refine from the local mode-discovery curve $C_{\text{seen},D}(T)$ and the empirical rank-frequency distribution. Available evidence is consistent with ≈ 50 named interventions covering the head of the per-hard-token failure distribution in many measured domains, but this number is a planning prior, not a universal constant: the same base model in cardiology RAG, legal contract drafting, and code review yields three different libraries because C_D, A_D, σ_D all change with the deployment domain.

Per-hard-token vs. sequence-level targets matter for SLA design. Per-hard-token residual error—the right metric for continuous-correction systems—has a polylog budget. Sequence-level failure probability—the right metric for one-shot systems—is strictly stricter and approaches full-catalogue coverage as k grows. Production deployments should design SLAs to match the actual cost structure, not read the per-token result as the production SLA.

Library design as engineering, not asymptotic theory. An intervention library can be assembled module-by-module, in any order, as long as modules target distinct failure classes. Layer-separated interventions (constrained decoding + retrieval + process supervision + tool call) compose approximately additively in our data; the Le [2026] caveat applies only to interventions sharing a prompt channel. Library construction is therefore highly parallelisable.

Capability libraries are coarser than error-class libraries. Five named math classes (arithmetic, units, counting, list manipulation, date arithmetic) collapse under one Python interpreter; three code classes (SyntaxError, NameError, most TypeError) collapse under one execution-feedback loop; format violations and the structural component of “missing required element”—together more than 20% of ErrorAtlas—collapse under one constrained-decoder deployment. A team budgeting for ≈ 50 interventions per domain can expect to need fewer than 50 capability-axis interventions while covering ≈ 50 named clusters. The six axes of Appendix B are closer to the right unit of engineering accounting than the twelve clusters.

6 Discussion

By-construction elimination as a boundary case. Seven of the 28 capability-elimination citations (Appendix B; constrained decoders [Suresh et al., 2025, Dong et al., 2025, Li et al., 2026, Zhang et al., 2023, OpenAI, 2024], proof kernels [Ren et al., 2025], static syntax checks) achieve residual error rate equal to zero *by mathematical construction*, not statistically. Constrained decoders set $P(\text{invalid token}) = 0$ at every generation step; the class of grammar-violating outputs is mathematically empty. In this regime, the polylog bound of Proposition 1 is loose: covered clusters contribute zero to residual error, not a small positive quantity. This is a strengthening result, not a counter-example—capability provisioning erases the error class entirely. Pattern A applies precisely to structural/verifiable classes (format, syntax, schema, invalid proofs), which are also the head of the heavy-tailed cluster frequency distribution. The strongest mechanism applies exactly where the catalogue is densest, compounding the polylog conclusion.

Counter-evidence relocates, not dissolves. Five prominent steep-decay papers [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026, Karpinska et al., 2024] are routinely cited as exponential-compounding evidence. Every one decays over a variable distinct from raw token length: compositional graph size [Dziri et al., 2023], number of supporting facts [Kuratov et al., 2024], log-time horizon [Kwa et al., 2025], capacity threshold [Wan et al., 2026], evidence scope [Karpinska et al., 2024]. The apparent rapid decay is, in each case, in a quantity our framework already concentrates the action in (k_{hard} and $|C|$), not in raw sequence length—a relocation, not a dissolution. These regimes remain hard; the framework’s value is directing intervention toward capability provisioning along the actual decay axis rather than toward context-window expansion that does not help. Appendix C walks through each paper.

Limitations. Postulate 1 is empirical, not derived; a domain with genuinely Heaps-power-law mode discovery would invalidate the doubly-logarithmic special case while preserving the qualitative polylog conclusion. The coverage form of §3.3 is an empirical best-fit; readers should not interpret the numerical match against anchors as theoretical confirmation. Cluster-orthogonality is approximate; Le [2026]’s caveat on prompt-channel interference tightens the bound in shared-channel settings. Domain narrowness: Postulate 1’s empirical anchor rests on three taxonomies (ErrorAtlas, HumanEval, MWPEs), all published 2025–2026 and covering general/code/math; the framework is untested on agentic workflows, long-running scientific reasoning, and multi-turn tool use over millions of tokens—precisely the regimes where $|C|$ may grow faster than logarithmically. The $\sigma \approx 1.85$ estimate is a single-point calibration, not a discovery-curve fit; a subsample-vs-distinct-modes plot has not been published for any LLM error taxonomy at this writing. β is latent and we report no empirical range. Taxonomy granularity (the L2–L3 gap) is unmeasured. Patch-shift between deployments changes $A_D, \sigma_D, \beta_D, |C_D|$ simultaneously; libraries calibrated against one patch under-cover the next. Finally, the framework *relocates* the difficulty of long-context reliability rather than resolving it: where k_{hard} grows with task length (adversarial compositional structure, multi-hop chains, long agent horizons), reliability remains hard. The contribution is to name the on-axis intervention, not to make those regimes easy. A falsifiability test the framework passes: DebugBench [Tian et al., 2024] explicitly catalogues which classes execution-feedback can and cannot repair, and the empirical split (works for syntax/reference; “unhelpful for logic errors”) is exactly what the framework’s selectivity claim predicts.

7 Conclusion

LLM reliability is often framed as an asymptotic scaling problem. Prior work in this line [Anonymous, 2025] argued that the framing rests on a false uniformity assumption—errors concentrate at ≈ 5 –10% of tokens. This paper takes the next step: within that sparse set, errors are not only concentrated but also *repetitive*, clustering into a finite catalogue whose size, under the empirical postulate of §3.2, grows logarithmically—and under the conservative Heaps alternative (Appendix A), as a small power—of observed failures. The conditional consequence (Proposition 1) is that the intervention budget required to bound per-hard-token residual error scales polylogarithmically in sequence length within a fixed domain patch, and becomes a domain-constant once the patch ceiling $|C_D|$ is reached. Sequence-level reliability targets are strictly tighter. Available evidence is consistent with libraries on the order of tens of interventions covering the head of the per-hard-token failure distribution in many fixed domains. The exact number is patch-indexed and should be estimated from local discovery and rank-coverage curves, not assumed from cross-domain priors. This reframes the question from “can we bound the growth of n -token error?” to “have we catalogued enough failure modes?” The latter is finite and addressable. Direct empirical measurement of the mode-rate constant σ on new domains—agentic, scientific, code-in-production—is the open empirical question this paper invites; future work flagged by Anonymous [2025, §6] on attention-mechanism bounds and stratified-manifold geometry would replace the empirical postulate with a derived result.

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544 A Heaps Power-Law Variant and Cluster-Count Sensitivity

545 A reader who prefers to derive cluster-count growth from standard Heaps’ law rather than from
546 Postulate 1 obtains a qualitatively similar result. Let $|C|(k_{\text{hard}}) \approx K \cdot k_{\text{hard}}^b$ with $b \in (0, 1)$. Canonical
547 fits give $b \approx 0.5$ for natural-language vocabularies; failure-mode taxonomies plateau much more
548 sharply (ErrorAtlas stabilises at $|C| = 17$ across 10^4 + failures), implying a smaller effective $b \approx 0.2$
549 in our setting. Composing with $k = \Theta(\log n)$, we get $|C| = O((\log n)^b)$, and Proposition 1 becomes

$$m = O\left((\log n)^{b \cdot (1 - \varepsilon / e_{\text{hard}})}\right).$$

550 This is still polylogarithmic in n for any $b \in (0, 1)$ and any $\varepsilon < e_{\text{hard}}$. The paper’s qualitative claim
551 survives either choice of cluster-count law.

552 **Symbolic-form sensitivity across candidate laws.** The polylog conclusion depends on which
 553 cluster-count law one accepts; available evidence is consistent with multiple candidates because no
 554 subsample-discovery curve has been published for any LLM failure-mode taxonomy at this writing.
 555 We therefore report symbolic rates rather than fitted constants. With $h(n) = \beta k(n)$:

- 556 • **Logarithmic:** $|C_{\text{active},D}(n)| = O(\log h(n))$. If $k(n) = \Theta(\log n)$, then $m =$
 557 $O((\log \log n)^{1-\varepsilon/e_{\text{hard}}})$.
- 558 • **Heaps:** $|C_{\text{active},D}(n)| = O(h(n)^b)$ with $b \in (0, 1)$. If $k(n) = \Theta(\log n)$, then $m =$
 559 $O((\log n)^{b(1-\varepsilon/e_{\text{hard}})})$.
- 560 • **Saturating:** $|C_{\text{active},D}(n)| \leq |C_D|$. Then $m = O(|C_D|^{1-\varepsilon/e_{\text{hard}}})$, constant in n once the
 561 patch ceiling is reached.

562 The qualitative conclusion is robust **under the $k(n) = \Theta(\log n)$ regime**: m grows more slowly than
 563 any positive power of n under every candidate, and the directional claim (“a small library covers the
 564 head of the failure distribution in the per-hard-token regime”) survives. Only the exponent shifts:
 565 doubly-logarithmic under logarithmic discovery, $(\log n)^b$ with small b under Heaps, constant in the
 566 cap regime. The doubly-logarithmic rate is the optimistic special case. If $k(n)$ grows as a positive
 567 power of n , the Heaps variant inherits that power and the polylog-in- n language fails along that
 568 axis—the framework’s intervention prescription still applies, but its asymptotic-rate framing does not.

569 Numerical constants require a measured discovery curve $C_{\text{seen},D}(T)$ or $C_{\text{active},D}(n)$. Existing tax-
 570 onomies provide endpoint category counts at a single corpus scale (ErrorAtlas at $|C| = 17$ for
 571 $\approx 10^4$ failures), not discovery curves. The subsample-discovery measurement remains the explicit
 572 empirical test that would either tighten the postulate or fall back to the Heaps variant. Until that
 573 measurement exists, the framework’s headline rate should be read as *polylogarithmic in the pre-cap*
 574 *regime, domain-constant in the cap regime*—with the specific exponent flagged as a falsifiability test
 575 rather than a fitted prediction.

576 B Full Failure-Mode Taxonomy and the Capability-Elimination Harvest

577 The intervention literature provides at least one targeted countermeasure for each named cluster of §4.
 578 Two organising granularities exist: at the capability level (six axes) the cluster-orthogonality property
 579 is most clearly visible; at the error-class level (twelve named clusters) the evidence aligns with the
 580 taxonomies of §4 but is finer-grained than the underlying capability mechanisms.

581 **Six capability axes and three structural patterns.** A dedicated harvest yields **28 quantitatively-**
 582 **anchored citations** across **six independent capability axes**: Arithmetic (Python/symbolic execution),
 583 Code Execution (REPL/sandbox feedback), Format/Structure (constrained decoding, FSMs, grammar
 584 engines), Perception/Grounding (visual grounding for GUI, charts, tables), Knowledge/RAG (dense
 585 retrieval and citation grounding), Verification (proof checkers, learned verifiers, classifier rerouting,
 586 process supervision). Each axis is independently confirmed by between three and nine citations. We
 587 stratify the 28 by kind of evidence into three patterns:

588 **Pattern A — Hard guarantees (by-construction).** Seven citations achieve residual error rate
 589 equal to zero *by construction*, restricted strictly to structural/verifiable classes: constrained decoders
 590 set $P(\text{invalid token}) = 0$ at every step [Suresh et al., 2025, Zhang et al., 2023, Dong et al., 2025,
 591 Li et al., 2026, OpenAI, 2024]; static syntax checks reject programs with a `SyntaxError` before
 592 execution [Wen et al., 2024]; proof kernels reject any output failing type-checking [Ren et al., 2025].
 593 The class of grammar-violating outputs is mathematically empty under these mechanisms.

594 **Pattern B — Strong empirical reductions with class-shift signature.** Roughly fourteen citations
 595 report empirical reductions of 80–100% in a named error class, with the post-intervention failure
 596 log dominated by structurally different residual classes. Program-of-Thoughts on GSM8K [Chen
 597 et al., 2023]: calculation errors drop from 30% of failures to 0%, residuals are 62% reasoning
 598 + 36% misunderstanding. OpenMedCalc [Goodell et al., 2025]: “only interpretation errors were
 599 identified.” Acurai [Wood and Forbes, 2024]: 100% hallucination elimination on RAGTruth (95% CI
 600 91–100%)—strong empirical, not by-construction.

601 **Pattern C — Moderate reductions (60–80%) with residuals shifting outside the target class.**
 602 Legal RAG [Dantart, 2026]: fabricated citations $> 30\% \rightarrow < 0.2\%$. GPT-5 SimpleQA with web

access [OpenAI, 2025]: 47% $\rightarrow$ 9.6% (inter-condition, not within-condition). CRITIC [Gou et al., 2024a]: toxic generation -79.2% .

The 28 citations, organised by axis and pattern.

Axis	Citation & targeted class	Pre $\rightarrow$ Post	Pattern
Verification	Ren et al. [2025] — Invalid Lean proof	any $\rightarrow$ 0% by constr.	A
Format	OpenAI [2024] — JSON schema violation	$>60\% \rightarrow 0\%$ by constr.	A
Format	Zhang et al. [2023] — Tool-call syntax	21–100% $\rightarrow 0\%$	A
Format	Suresh et al. [2025] — JSON parse failure	13–82% $\rightarrow 0\%$	A
Format	Dong et al. [2025] — Multi-format errors	20–38% $\rightarrow 0\%$	A
Format	Li et al. [2026] — Malformed tool calls	33–78% $\rightarrow 0\%$	A
Arithmetic	Chen et al. [2023] — Calculation on GSM8K	30% $\rightarrow 0\%$ of failures	B
Arithmetic	Goodell et al. [2025] — Clinical arithmetic	“only interpretation errors”	B
Knowledge/RAG	Wada et al. [2025] — RAG hallucinations	8% $\rightarrow 0\%$ ($p = 0.012$)	B
Knowledge/RAG	Wood and Forbes [2024] — Context-conflict hallucinations	100% $\rightarrow 0\%$ on subset	B
Knowledge/RAG	Dantart [2026] — Fabricated legal citations	$>30\% \rightarrow <0.2\%$	C
Code Exec	Wen et al. [2024] — <code>SyntaxError</code> (HumanEval)	5.76% $\rightarrow 0.01\%$	A
Code Exec	Li et al. [2022] — False-positive submissions	62% $\rightarrow 4\%$	B
Code Exec	Shi et al. [2024] — 5 code-bug classes	100% repair (5/6 classes)	B
Knowledge/RAG	Gao et al. [2023b] — Citation grounding (ELI5)	$\approx 50\%$ w/o complete support	C
Knowledge/RAG	OpenAI [2025] — Factual errors	47% $\rightarrow 9.6\%$ w/web	C
Knowledge/RAG	Zakka et al. [2024] — Clinical citation errors	44% $\rightarrow 9\%$	C
Arithmetic	Wang et al. [2025a] — Arithmetic in medical reasoning	426 $\rightarrow$ 74 errors	B
Code Exec	Wen et al. [2024] — <code>NameError</code> repair	22.7% $\rightarrow 2.3\%$	C
Perception	Gou et al. [2025] — GUI grounding errors	16.2% $\rightarrow 73.3\%$ acc.	B
Perception	Xie et al. [2025] — GUI task failures	5% $\rightarrow 27\%$ SR ($5.4\times$)	B
Perception	Cheng et al. [2024] — Element mislocation	5.2% $\rightarrow 53.4\%$ ($10.3\times$)	B
Perception	Liu et al. [2023] — Chart-reading errors	38.2% $\rightarrow 67.6\%$ (+29.4 pp)	B
Verification	Gou et al. [2024a] — Toxic generation	-79.2%	C
Verification	Wang et al. [2024b] — Reasoning step errors	28.6% $\rightarrow 43.5\%$ w/rerank	B
Knowledge/RAG	Asai et al. [2024] — Unsupported facts	55.5% $\rightarrow 22\%$ error	B
Perception	Lu et al. [2024] — Icon/element errors	70.5% $\rightarrow 93.8\%$ (79% red.)	B
Knowledge/RAG	Mallen et al. [2023] — Long-tail entity errors	$\approx 80\% \rightarrow \approx 50\%$ failure	B

Capabilities are coarser than error classes. Several entries in the harvest reveal “two-for-one” reductions where one capability addresses multiple named clusters: a single Python interpreter removes

the execution-error component of arithmetic, unit conversion, simple counting, list manipulation, and date arithmetic; constrained decoding eliminates by construction both format violations and the structural component of “missing required element”; code execution feedback strongly reduces `SyntaxError`, `NameError`, and most `TypeError` together; RAG strongly reduces factual hallucinations, fabricated citations, and outdated information jointly. The practical capability library required is therefore smaller than the count of named error categories.

The twelve named clusters.

Cluster	Axis	Intervention	Before → After	Patt.
A. Arithmetic	Arithmetic	PAL Python [Gao et al., 2023a]	20.1% → 61.5%	B
B. Unit conversion	Arithmetic	Folded into A (Python)	—	—
C. Counting	Arithmetic	Explicit counter (gap)	—	gap
D. Format/schema	Format	DINGO/SLOT [Suresh et al., 2025, Wang et al., 2025b]	up to +68 pp; 99.5% acc.	A
E. Code logic/type	Code Exec	Reflexion/AgentCoder [Shinn et al., 2023, Huang et al., 2023]	80% → 96.3% pass@1	A/B
F. Multi-hop drift	Knowledge	Entity-grounded rewriter	≈5–10 pp	C
G. Reasoning step	Verification	Math-Shepherd [Wang et al., 2024b]	28.6% → 43.5%	B
H. Spec misinterpret.	Verification	Clarification [Niwa and Iso, 2024]	ROUGE-L +15	C
I.a Struct. missing	Format	Subsumed by D (constr. decode)	—	A
I.b Semantic missing	Verification	Subsumed by G/H	—	C
J. Hallucination	Knowledge	RAG [Wood and Forbes, 2024]	non-zero → 0%	B/C
K. Refusal	Verification	POROver [Karaman et al., 2024]	57.6% → 82.1%	B
L. Tool/API	Format+Verif.	SAGE-Agent [Suri et al., 2025]	36.5% → 65.2%	B

Eight of twelve categories have a strong citation with double-digit percentage-point improvement; three (B, C, I) lack a clean single-cluster ablation; one (F) has consistent medium-strength evidence. Category I (missing required elements) splits mechanistically into I.a (structural, absorbed by D’s constrained decoders) and I.b (semantic, absorbed by G/H). Categories B (units) and C (counting) remain technical gaps: B is naturally folded into A; C is the smallest residual gap.

Additivity and its limits. Patel et al. [2026] demonstrate that stacking interventions targeting orthogonal failure modes can produce large compound reliability gains: their parallel-consensus framework yields a $14,700\times$ improvement over single-pass baseline—evidence for compound gains from decomposition plus consensus aggregation under their specific parallel-voting regime, not direct demonstration that heterogeneous cluster-targeted interventions compose additively without voting. Shang et al. [2024] similarly show supra-additive gains when modules target distinct failure modes. Le [2026] reports that schema-level and prompt-level instructions interact *non-additively* when sharing a prompt channel: interventions on orthogonal processing layers (decoding constraint vs. retrieval vs. training-signal vs. inference-time tool call) compose near-additively, while those sharing a channel may interfere.

The irreducible-semantic residual. Of the 17 ErrorAtlas categories, 13 are addressed under Patterns A, B, or C by one of the six capability axes; four are not—residuals of inappropriate refusal beyond preference optimisation, specification misinterpretation not closed by clarification, problem-decomposition reasoning bottlenecks, and a “user wanted something different” semantic remainder. These are classes where the failure is in choosing what to do, not executing it. Proposition 1’s prediction of $O(\log)$ rather than $O(0)$ residual reliability reflects exactly this irreducible core. The framework does not claim 100% coverage of all failures; it claims polylog-bounded *capability-eliminable* failures, with the named semantic residual as the floor.

C Counter-Evidence Re-Audits

Five prominent papers are routinely cited as evidence that LLM reliability decays steeply with length [Dziri et al., 2023, Kuratov et al., 2024, Kwa et al., 2025, Wan et al., 2026, Karpinska et al., 2024]. A careful re-reading shows that *every one* of these papers decays over a variable distinct from raw token length n . Identifying the decay axis is not the same as dissolving the practical concern: where compositional graph size, fact count, or evidence scope grow with problem length, the framework predicts steep failure curves. The contribution is to identify which interventions help (capability provisioning along the actual decay axis) and which do not.

Dziri et al. [2023], Faith and Fate. GPT-4 multi-digit multiplication accuracy drops from 59% (3-digit) to 4% (4-digit) zero-shot, with the authors theorising “probability of incorrect predictions converges exponentially to ≈ 1 for abstract compositional tasks.” The decay variable is *compositional graph size* N , not raw token length: the 3×3 graph has on the order of d^2 partial products plus carries, and multi-digit multiplication is engineered so that $k_{\text{hard}} \approx N$ —every node in the computation graph is a hard decision. For natural-language tasks where $k_{\text{hard}} \ll n$, Dziri’s regime is the *boundary case* in which our framework reduces to their result. A clean $(1 - \varepsilon)^N$ exponential cannot simultaneously reproduce 59% at 3-digit and 4% at 4-digit for any single per-node ε —the observed drop is locally steeper than per-node iid exponential, consistent with a finite catalogue of failure modes exhausting as N grows. *Where the framework concedes:* in adversarial compositional tasks engineered so $k_{\text{hard}} \approx N$, the framework reduces to Dziri’s regime and does not relieve it; the relocation is informational, not magical.

Kuratov et al. [2024]. The abstract reports “performance declines sharply with increased reasoning complexity” and models “effectively utilise only 10–20% of the context.” Read in detail, BABILong varies two axes: context length n (0K to 10M tokens) and number of supporting facts k (QA1 = 1 fact to QA3 = 3 facts). The sharp decline is in k , not n . For QA1 (single-fact), most models “perform well up to 4,000 tokens”—a plateau, not exponential decay. The famous RAG-flat-across-length result (60% on single-fact QA *independent of context length*) is the cleanest possible demonstration that when relevant evidence is in window, length does not matter. Recurrent Memory Transformers maintaining performance to 50M tokens further confirms effective k is determined by architecture, not raw n . The framework’s concession is narrow but real: for multi-hop tasks whose required fact count grows with task complexity, BABILong’s sharp k -axis decay is exactly what the framework predicts happens, not a counter-example.

Kwa et al. [2025] (METR). Per-model success fits a logistic in $\log(\text{human-task-duration})$: $S(t) = \sigma(\beta(\log t - \log h))$. Logistic-in-log-length is *mathematically sublinear* in length itself. The 80%-horizon being 4–6 $\times$ shorter than the 50%-horizon is a steep within-model cliff but compatible by construction with the polylog result—a cliff at a specific capacity threshold is a manifold-transition signature. The famous exponential—capability doubling every seven months—is an *inter-model* claim about how the horizon h moves across generations, orthogonal to within-model decay shape. An honest qualifier: the within-model logistic-in-log cliff is steep on a practical scale. Calling it “sublinear in length” is technically correct but engineering-useful only for systems designed to operate well below the 50% horizon.

Wan et al. [2026], Fano-style upper bound. This paper theorises super-linear information-demand growth and identifies an “accuracy cliff” at capacity overflow: “when the task’s information demand surpasses the model’s output capacity, performance does not degrade gracefully but instead collapses sharply.” The behavioural prediction is a *threshold*, not smooth $(1 - \varepsilon)^n$. A cliff at a specific capacity threshold is exactly the manifold-transition behaviour our framework predicts at the boundary between covered and uncovered modes; Wan et al.’s mechanism (capacity overflow) is one specific cause, complementary to ours. *Where the framework concedes:* Wan documents a mechanism by which $|C|$ effectively exceeds the model’s coverage in a single forward pass; Pattern A interventions (constrained decoding, formal verification) are the kind of response the framework expects but does not automatically supply.

Karpinska et al. [2024]. GPT-4o achieves 55.8% pair accuracy across 1,001 minimally-different true/false claim pairs about long fictional books (mean length 127K tokens). The paper reports performance by *evidence scope*, not by context length: 59.8% on sentence-level retrieval, 47.6%

694 on passage-level, 41.6% on global reasoning. No per-context-length curve is reported. The decay
 695 axis is the number of evidence pieces that must be integrated—a k -axis quantity, not an n -axis one.
 696 NoCha is therefore not counter-evidence to a sublinear-in- n claim. Evidence-scope decay is exactly
 697 the k -axis observation the framework predicts cannot be addressed by scaling raw context length—it
 698 requires retrieval or process supervision along the actual decay axis. The relocation is informational,
 699 not magical.

700 **Unifying observation.** Every counter-paper decays over a variable that is not n : compositional
 701 graph size, fact count, log-time horizon, capacity threshold, or evidence scope. The apparent rapid
 702 decay is, in each case, in a quantity our framework already concentrates the action in (k_{hard} and $|C|$),
 703 not in raw sequence length. This is a relocation, not a dissolution. Where k_{hard} grows with task
 704 length—adversarial compositional structure, multi-hop fact chains, long horizons that force more
 705 decisions—reliability remains hard; the framework’s value is directing intervention toward capability
 706 provisioning along the actual decay axis rather than toward context-window or compute-budget
 707 expansion that does not help.

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1079 scientific rigor, or originality of the research, declaration is not required.

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